Chapter I

SUMMARY OF THE PROJECT

A. Introduction

AeroGenesis is an innovative aerial combat simulation game designed to bring to life the dreams of aviation enthusiasts and aspiring aircraft creators. Many gamers and aviation fans imagine flying and designing their own aircraft and AeroGenesis turns that dream into reality. The game empowers players to engineer, customize, and pilot aircraft based on their imagination, creativity, and strategic vision. Unlike traditional flight combat games that rely on historically fixed aircraft, AeroGenesis integrates a realistic aerodynamic physics system that simulates the true behavior of lift, drag, thrust, and stability. Through engineering-based gameplay and a virtual wind tunnel testing feature, every aircraft must prove its flight-worthiness before entering combat. AeroGenesis combines creativity, engineering, and competitive multiplayer warfare — offering players the freedom to build the future of air combat while experiencing the thrill of dominating the skies with a plane that they designed. It invites players to dream, create, and conquer, one aircraft at a time. Aerogenesis is a modular aircraft design, simulation, and gameplay platform that integrates engineering-grade design tools with real-time flight simulation and interactive gaming environments. The platform is structured around two core modules: Creation Mode and Game Mode, enabling a full design-to-flight workflow.

B. Creation Mode

Creation Mode provides players with a complete set of tools to design and build their own aircraft from the ground up. In this mode, users can shape the structure, configure aerodynamic performance, integrate onboard systems, and customize the aircraft's appearance. The mode includes several categories of tools that guide players through the aircraft development process.

The Structural and Design Tools allow users to construct the main physical components of the aircraft. The Fuselage Builder enables players to shape the main body of the aircraft, while the Wing Designer allows the creation of wings with adjustable span, chord, and sweep. Additional tools are available for designing the aircraft's stability components, including the Tail and Stabilizer Tools for vertical and horizontal stabilizers. Players can also use the Control Surface Editor to add and configure ailerons, rudders, elevators, and flaps, and the Landing Gear Tools allow customization of landing gear type, placement, and retractable mechanisms.

The Aerodynamics and Performance Tools focus on improving flight characteristics and ensuring the aircraft functions properly in flight. Users can select or modify wing and tail profiles through the Airfoil Selector or Editor, while the Weight and Balance Calculator help ensure the aircraft's center of gravity is correctly positioned. More advanced analysis features are also available through integrated CFD (Computational Fluid Dynamics) and FEM (Finite Element Method) structural analysis tools, allowing users to evaluate aerodynamic performance and structural integrity.

The Systems and Electronics Tools allow players to design the operational aspects of the aircraft. The Cockpit and Instrument Designer enable customization of flight controls and displays, while the Control Systems Builder connects cockpit inputs to the aircraft's control surfaces. Players can also configure onboard electronics through the Avionics Setup, which includes navigation systems, communication equipment, and autopilot functions.

Finally, Aesthetic and Customization Tools allow users to personalize the appearance of their aircraft. The Paint and Livery Editor let players apply colors, decals, or camouflage patterns, while the Shape and Texture Editor enable adjustments to visual details for either realism or stylistic preference. Additional Accessory Tools allow users to add optional components such as weapons, sensors, or other external attachments when needed.

C. Game Mode

This mode presents players with three main options. In addition to providing ready-to-use aircraft, it also allows users to bring their own custom-designed aircraft into the game and test them in various gameplay scenarios. The simulation includes six degrees of freedom (6-DOF) flight model, enabling realistic motion in three translational and three rotational axes, which is essential for flight dynamics analysis, control system experimentation, pilot training, and aircraft performance evaluation.

Players can choose from several gameplay modes designed for different experiences. Acrobatic Mode focuses on racing, aerial stunts, and mastering complex maneuvers. Story Mode offers immersive missions and challenges

that test both piloting skills and aircraft design capabilities. Player vs. Player (PvP) Mode allows players to participate in intense 10-versus-10 aerial battles against other users. Additionally, matches can be played under two rule sets: Classic Mode, where strict physical realism is relaxed to emphasize accessibility and fun, and Rank Mode, where accurate physics and realism are prioritized to create a more competitive and simulation-focused environment.

Aero Genesis combines creativity, strategy, and action, making it the ultimate experience for aviation enthusiasts and gamers alike. It is a PC-based multiplayer aircraft design and flight simulation game that combines realistic aerodynamics, physics-based modeling, and immersive gameplay. Unlike traditional flight simulation games, AeroGenesis allows players to design aircraft from scratch, where engineering decisions such as wing shape, weight distribution, thrust, and structural design directly affect flight performance. The game applies real-world aerodynamic and structural principles, ensuring that aircraft behavior is determined by the quality of the user's design. The project features both offline and online multiplayer dogfights, dynamic weather conditions, and support for advanced control systems such as joysticks and virtual reality.

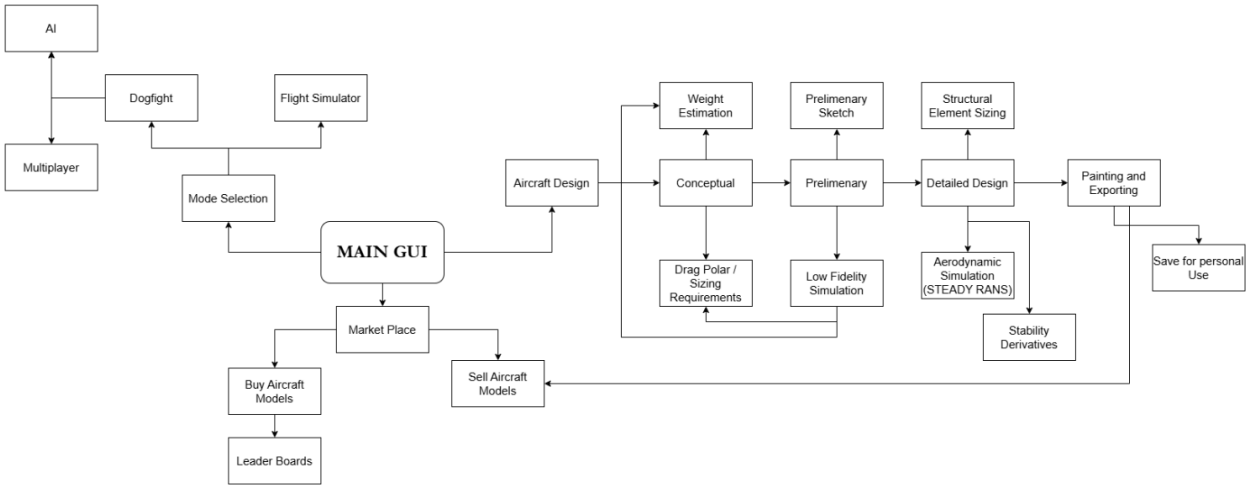


Figure 1.1 GUI Workflow

AeroGenesis also integrates engineering-grade tools, including aerodynamic, structural, and optimization simulations, supported by guided tutorials and AI-assisted learning. Additionally, a secure in-game marketplace enables players to buy and sell original aircraft designs, protected by anti-cheating and intellectual property safeguards. Designed for aviation enthusiasts, students, and flight simulation gamers, AeroGenesis aims to provide an innovative platform that merges education, creativity, and competitive gameplay while promoting a deeper understanding of aircraft design and flight mechanics.

D. Name of the Project

"AeroGenesis" literally means "the creation of flight" or "the birth of an aircraft." This perfectly aligns with the project's main feature allowing players to design original aircraft from the ground up, where performance is shaped by engineering decisions rather than presets.

"Aero" comes from aerodynamics and aviation, directly relating to aircraft, flight, and air-based engineering. "Genesis" means origin, creation, or beginning, symbolizing the act of designing and building something from scratch.

AeroGenesis stands out as a name that represents innovation, originality, and the engineering origins of flight, making it both memorable and meaningful.

E. Location Of The Project With Vicinity Map

We chose 149 J.P. Laurel Ave, Lanang, Davao City, 8000 Davao del Sur as a location for **"AeroGenesis"** headquarters because it places us at the heart of a vibrant community of aviation enthusiasts and gamers. By being here, we can directly engage with our target audience, build strong connections, and create experiences that resonate with them. The area also provides opportunities to host events, workshops, and collaborations that help grow our brand and establish ourselves as a hub for innovation in aviation and gaming. This strategic choice supports both our community focused approach and long-term growth objectives.



Figure 1.1 Location of The Project



Figure 1.2 Location of The Project

F. Market Share

Market demand and market supply consisting of academe and gamers

Market demand = 550,000

Market Supply = 40,000

$$\text{Market Gap Ratio} = \frac{550{,}000 - 40{,}000}{550{,}000}$$

$$\text{Market Gap Ratio} = \frac{510{,}000}{550{,}000}$$

$$\text{Market Gap Ratio} \approx 0.9273$$

Market Gap (%) = 0.9273 × 100 = **92.73%**

G. Market Projection

1. Aviation Enthusiasts & Hobbyists- People who are naturally passionate about aircraft, engineering, and flight.

2. Gamers (Mainstream & Niche) - Those who enjoy action games, simulation games, or competitive gameplay.

3. Creative Builders & Designers Players who love creating things.

4. Students (High School & College) Especially from STEM, engineering, and tech tracks.

5. Digital Entrepreneurs People who create digital assets and monetize them.

6. Local Defense & Tech Community While small, it's a serious and engaged niche.

Segment	Size	Interest Level	Revenue Potential
Aviation Enthusiasts & Hobbyists	Small–Medium	**30%**	**30% – 45%**
Gamers	Large	**30%**	**25% – 35%**
Creative Builders	Medium	**20%**	**25% – 35%**
Students	Large	**10%**	**10% – 15%**
Digital Entrepreneurs	Small	**5%**	**5% – 10%**
Defense Community	Very small	**5%**	**5% – 10%**

Table 1 Market Projection

H. MARKET PRICING

The pricing of our product is separated into two. By using similar UGC (User-Generated Content) companies as benchmark for this price we would be able to come up with a safe and reliable approach and adapt with the current trend.

Features/Services	**Pricing**
Game Flight Simulator/ VS mode	Free To Play
Aircraft Creation Mode	One Time Payment (₱499)
Addon Tools (For Optimization)	₱150 – ₱300
UGC Price Caps	
Light Aircraft Normal and Utility Category	₱ 50 – ₱60
Commercial Aircraft	120 – ₱250
Military Roles	
Interceptor	₱150 – ₱1000
Multi Role	₱300 – ₱1000
Fighter Jets	₱350 – ₱1000
Acrobatic Fighters (WW II)	₱120– ₱300
Stealth Bomber Aircraft	₱500 – ₱1000

Table 1.1 Product Cost

The reason each aircraft category has a different expected price in your UGC marketplace is because the **complexity of design**, **time investment**, and **player-perceived value** are not the same across the three types. Interceptors are generally the cheapest because they are the simplest to design: their airframes are basic, their avionics are minimal, and they perform mostly one role—high-speed interception. This means creators spend less time modeling them, balancing their physics, or setting up weapons and sensors, and therefore the expected market value stays lower.

Multirole fighters sit in the middle range because they are more advanced and require significantly more work. A creator must tune the aircraft for several missions—

dogfighting, strike missions, ground attack, and defense—while also building more detailed cockpits, weapon systems, and aerodynamic behavior. Because these planes are more universally useful in gameplay and take longer to build, players place a higher value on them.

Stealth aircraft naturally fall into the highest price bracket because of their extreme complexity. Their shapes must follow stealth geometry constraints, and they usually require internal weapon bays, advanced avionics, next-generation sensors, and very precise aerodynamic behavior. These features take much longer to model and simulate, and only skilled creators can build them properly. As a result, stealth aircraft become the premium category both in terms of design effort and marketplace price

I. UGC Cuts

Choosing a 70% creator share and 30% studio share places AeroGenesis among the most creator-friendly UGC ecosystems in the industry. While large companies like Valve, Epic, and Roblox typically take much bigger cuts, a higher share for creators gives your platform a strong competitive advantage. This model strongly encourages aircraft designers, students, and hobbyists to contribute high-quality content because they feel fairly rewarded for their effort and creativity. A 70/30 split also increases the volume of available aircraft in the marketplace, which boosts player engagement and keeps the in-game economy active. Meanwhile, the studio's 30% cut remains enough to cover essential costs such as server hosting, moderation, platform maintenance, and continued updates. By prioritizing creator earnings, our studio builds long-term loyalty, attracts skilled designers from the aviation community, and strengthens the project's overall sustainability.

J. Investment Cost

The pricing of our product is separated into two. By using similar UGC (User-Generated Content) companies as benchmark for this price we would be able to come up with a safe and reliable approach and adapt with the current trend.

Assets and Liabilities			
Machines & Equipment	**Quantity**	**Cost per Unit (₱)**	**Total Amount (₱)**
High-Spec PC (Dev Workstation)	11	75,000	825,000
Mid-Spec PC (Support/Admin)	4	35,000	140,000
Server (Dedicated)	1	180,000	180,000
Printer	1	15,000	15,000
Drawing Tablet	3	8,000	24,000
Split Type Aircon	2	40,000	80,000
Projector	1	20,000	20,000
Network Setup (Routers, Switches, Cabling)	1	35,000	35,000
Office Tables	4	9,000	36,000
Cubicles (2 per set)	5	13,000	65,000
Office Chairs	15	2,000	30,000
Conference Table	1	18,000	18,000
Conference Chairs	10	1,500	15,000
Whiteboard	1	7,000	7,000
Shelves	2	2,500	5,000
Office Steel Cabinet	2	9,500	19,000
Office Sofa Set	1	15,000	15,000
Business Permit & Registration	1	25,000	25,000
Software Licenses (Game Engine, Tools, Security)	1	150,000	150,000
Office Renovation (50 sqm)	1	1,200,000	1,200,000
Total			2,904,000

Table 1.2 Assets and Equipment

Development Phase			Post Development Phase		
Position	**Monthly Salary (₱)**	**Annual Salary (₱)**	Technical Lead / Maintenance Programmer	₱40,000	₱480,000
Lead Engineer	₱70,000	₱840,000	Server / Backend Maintenance	₱35,000	₱420,000
Gameplay Physics Programmer	₱55,000	₱660,000	Content / 3D Artist (updates & fixes)	₱28,000	₱336,000
Game Mechanics Programmer	₱50,000	₱600,000	QA / Community Support	₱20,000	₱240,000
Backend / Server Programmer	₱60,000	₱720,000	Marketing / Community Manager	₱22,000	₱264,000
Aircraft Modeler Developer	₱45,000	₱540,000	Admin / Operations	₱18,000	₱216,000
3D Modeler	₱35,000	₱420,000	Total	₱163,000	₱1,956,000
UI/UX Developer	₱35,000	₱420,000			
Environment / Effects Artist	₱38,000	₱456,000			
QA / Support Team	₱25,000	₱300,000			
Marketing	₱30,000	₱360,000			
Admin	₱25,000	₱300,000			
Total	₱468,000	₱5,616,000			

Table 1.3 Employee salary during development and Post Development Phase

Permits and License Agreements	Cost
Steam Cut	20% of the Revenue
Unreal Engine	3.5% of the Gross Revenue after hitting $1Million (since We will Upload it to EPIC games)
Business Permit	25,000

Table 1.4 Permit and License Cost

K. Market Financing

For the early phase, which is the prototype phase, our main focus would be to establish the system architecture for the early access of our project. We will be inviting investors to invest in or game. To support the development, market entry, and commercialization of the project, a market profit–sharing investment model is proposed. This financial approach is structured to attract external investors while maintaining full operational and technical control within the project development team. The model emphasizes transparency, sustainability, and equitable return based on market performance.

L. Purpose of Investment

The investment funds will be utilized specifically to strengthen the project's market position and commercial viability through the following areas:

- Product development and system enhancement
- Market deployment, promotion, and customer acquisition
- Software optimization and technical support

- Operational infrastructure and licensing requirements
- Continuous improvement aligned with market demand

Under this model, investors shall be entitled to 10% of the market profit generated by the project.

Definition of Market Profit- Market profit refers to the net income derived directly from market-related activities, including product sales partnerships, after deducting market-specific expenses such as marketing costs, distribution fees, and sales operations.

M. Financial Transparency and Risk Management

To ensure accountability and investor confidence, the following measures shall be implemented:

- Regular market performance and financial reports
- Clearly defined accounting methods for market profit calculation
- Controlled and trackable marketing and sales expenditures
- Periodic evaluation of market trends and risks

N. Market Feasibility (Local – Philippines)

The potential audience for an aircraft design and simulation game in the Philippines can be estimated by examining several relevant groups of users. The largest group consists of PC gamers, which is estimated to range between 5 and 7 million individuals. Within this population are players who are already

familiar with simulation-style games and may be open to trying a new title that focuses on aircraft design and flight dynamics. In addition to the general gaming population, there is a smaller but highly relevant group of aviation hobbyists and enthusiasts, estimated to number around 20,000 to 40,000 people. These individuals typically have a strong interest in aircraft technology, flight mechanics, and realistic simulation environments. Another potential audience includes students in STEM and engineering-related programs, estimated at 200,000 to 300,000 individuals, many of whom may find value in a platform that combines gameplay with concepts related to aerodynamics, aircraft structures, and flight performance.

Considering the specialized nature of an aircraft design and simulation game, only a portion of the broader gaming population is likely to become active players. A reasonable estimate suggests that about 1% to 3% of PC gamers may be interested in this type of experience. This would result in a potential player base of approximately 50,000 to 200,000 individuals who may be interested in trying the game due to its combination of aircraft design, simulation, and competitive gameplay.

For a newly developed indie game, the initial number of players captured will likely be smaller and will grow gradually over time. In the first year, the game may attract roughly 250 to 2,000 players, primarily consisting of early adopters, aviation enthusiasts, and simulation fans who actively search for niche flight-related experiences. As the game improves through updates, community engagement, and word-of-mouth promotion, the second year may see growth to around 500 to 6,000 players. If development continues successfully and the

game gains stronger recognition within relevant communities, the third year could see the player base expand to approximately 1,500 to 16,000 players.

When examining the expected composition of players, aviation enthusiasts are likely to represent the largest share, accounting for approximately 30% to 45% of the player base, as they tend to value realistic flight mechanics and aircraft design features. Simulation-oriented PC gamers, including players who enjoy games such as flight simulators or vehicle simulation titles, may account for around 25% to 40% of users. Another group consists of creative or builder-type players, estimated at 10% to 25%, who are primarily interested in designing and experimenting with aircraft rather than focusing solely on combat or competitive gameplay. Lastly, engineering and STEM students may represent roughly 5% to 15% of the player base, as the game may also attract individuals who are curious about aerodynamics, aircraft structures, and engineering-related concepts.

O. Technical Feasibility

The pricing of our product is separated into two. By using similar UGC (User-Generated Content) companies as benchmark for this price we would be able to come up with a safe and reliable approach and adapt with the current trend.

1. Required Core Technologies
    a. Game Engine – Unreal Engine 5
    b. Physics 6 DOF Simulation – JSBSIM

c. Multiplayer Back end – Use Cloud Services Like AWS LightSail

d. User Content Creation System – Needs A dedicated Team

e. Anti IP violation system – Relies on Shape Detection algorithm and Player Reports

2. System Architecture

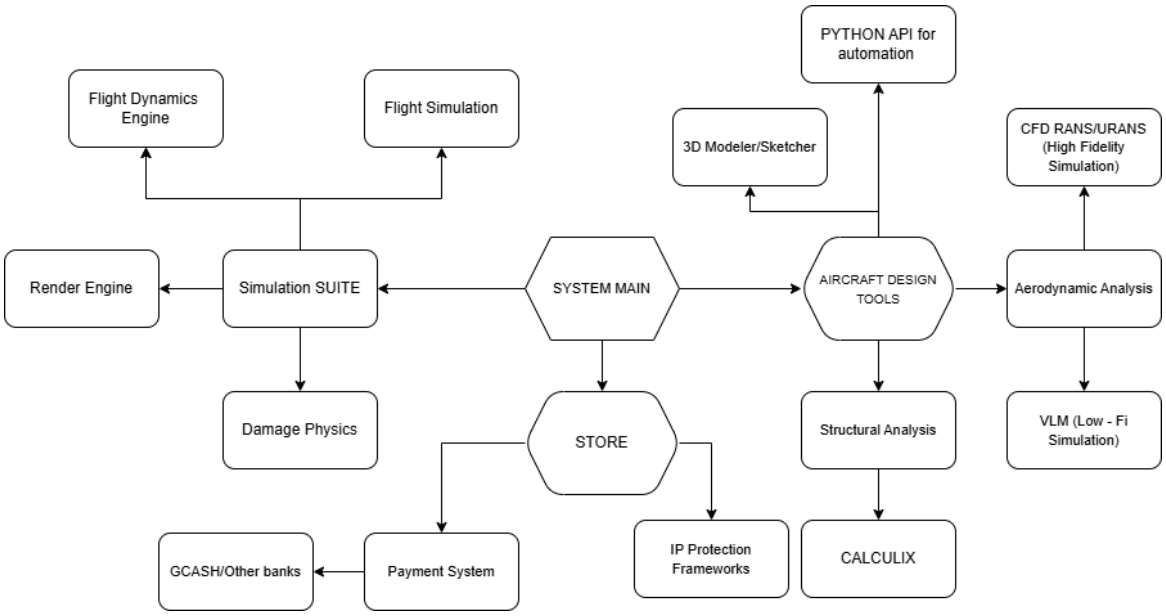


Figure 1.3 System Architecture

3. Human Resource Feasibility

The Philippines has a strong and growing talent pool in software engineering, game development, 3D modeling, and simulation-related disciplines. Local universities consistently produce graduates with backgrounds in computer science, engineering, and digital arts, while the outsourcing and

game industries have already built a workforce familiar with real-time engines, physics programming, and pipeline-based development

4. Hardware & Infrastructure Feasibility

The required hardware—development workstations, GPU-equipped systems, servers for builds and backend services—is commercially available within the country. Local suppliers and distributors can support procurement and maintenance.

5. Software Feasibility

The project benefits significantly from the availability of mature, industry-grade software tools:

- Unreal Engine provides a robust real-time rendering, physics, and tooling ecosystem.
- JSBSim offers a proven open-source flight dynamics model suitable for realistic aircraft simulation.

Both platforms are free and widely supported, allowing the team to benchmark architecture early, rapidly prototype and iterate and Reduce development risk from low-level technical uncertainties

The real challenge is not access to software, but integration complexity—bridging game logic, physics fidelity, and user-generated content without creating technical debt. Tools won't slow you down; bad architecture will.

6. Development Timeline

The development timeline is highly sensitive to the composition, experience, and technical maturity of the development team. Progress depends heavily on familiarity with Unreal Engine internals, competence in physics and simulation programming, and the ability to design scalable systems that can support user-generated content and modding without becoming unstable or unmaintainable. While an inexperienced team can still complete the project, this typically results in longer iteration cycles, increased refactoring, and delayed feature maturity. A realistic development schedule must therefore account for training and ramp-up periods, toolchain stabilization, and repeated validation of physics and gameplay systems. In practice, timelines can be adjusted—but technical competency cannot be bypassed without consequences.

P. Overall technical feasibility

With the currently available technology, development tools, and local resources, the project is technically feasible, and there are no fundamental engineering limitations that prevent its implementation. However, this feasibility is conditional and accompanied by significant caveats. The inclusion of user-generated content increases system complexity and places additional demands on validation, stability, and security. The quality and specialization of manpower directly influence both the development timeline and long-term maintainability of the system. Hardware costs must be actively managed due to market volatility, and development schedules must be grounded in actual team capabilities rather than optimistic assumptions.

Chapter II

MANAGEMENT ASPECT OF THE PROJECT

A. Pre-Operating Period

AeroGenesis plans to establish a company that develops an aircraft design and flight simulation platform integrating engineering-based aircraft creation with interactive simulation and gameplay modes such as aerobatics, racing, player-versus-player (PVP), and story-driven missions. An eighteen-month pre-operating period will be dedicated to organizational planning, system development, and resource preparation.

During this phase, the team will develop the core simulation engine, aircraft design tools, flight physics systems, and game mode frameworks, with support from technical consultants as needed. Management will finalize business policies, operational strategies, and development goals, while also formulating marketing plans, pricing models, distribution strategies, and potential industry and academic partnerships.

The proponents will assess technical feasibility, required technologies, workforce needs, and market demand to ensure the platform meets educational and industry standards while appealing to aviation enthusiasts and students. A lean organizational structure will be maintained to control costs, with additional staff hired upon full operation.

By the end of the pre-operating period, AeroGenesis will be fully registered, structured, and ready for full-scale operations.

B. Logo Of AeroGenesis Roofing Solutions

Figure 2.1 Logo Of AeroGenesis Roofing Solutions

C. Organizational Chart



Figure 2.2 Organizational Chart

D. The Operating Period

Upon completion of the eighteen-month pre-operating period, AeroGenesis will begin full-scale operations centered on its aircraft design and flight simulation platform that integrates engineering-based aircraft creation with interactive simulation. The operating period will focus on delivering a fully functional system that allows users to design, analyze, and fly custom aircraft within a realistic simulation environment. Key activities will include continuous enhancement of the aircraft design modules, flight engineering systems, and simulation accuracy to ensure alignment with real-world engineering principles. Interactive gameplay modes such as aerobatics, racing, player-versus-player (PVP), and story-driven missions will be fully deployed and regularly updated to enhance user engagement. The company will manage platform operations through system maintenance, performance optimization, content updates, and customer support services. Marketing initiatives, pricing implementation, and distribution strategies will be executed to expand the user base, while partnerships with academic institutions and industry organizations will support educational use and technical relevance. During this period, AeroGenesis will scale its workforce and resources in response to operational demands, prioritizing innovation, reliability, and long-term sustainability.

E. Type Of Business Organization

The type of business organization for AeroGenesis will be a Line and Staff structure. In this setup, authority flows from top management to the operational levels. The CEO and COO will serve as partners, working closely together to guide the strategic direction and day-to-day operations of the company. The CEO will oversee overall strategy, major business decisions, and long-term planning, while the COO will translate these plans into actionable operations and coordinate the activities of the company's departments. Reporting directly to the COO, the Lead Software Engineer will be responsible for developing and maintaining the AeroGenesis platform, ensuring that all software design and simulation tools meet high performance and quality standards. The Aerodynamics and Flight Simulation Specialists will handle aircraft modeling, testing, and optimization, while Marketing and Community Managers will manage promotions, user engagement, and educational partnerships. Administrative staff will maintain office operations and provide general support to ensure smooth and efficient workflow. This structure combines partnership at the top with a clear chain of command below, promoting collaboration, accountability, and alignment with the company's goals.

F. Form Of Ownership

AeroGenesis is an innovative aviation technology and flight simulation company that will provide a platform for aircraft design, development, and testing. The company will operate under a partnership structure, with the CEO and COO serving as partners who share responsibility, control, and profit

distribution. The business will be legally registered with the Securities and Exchange Commission (SEC) and other relevant government agencies, ensuring compliance with regulations, taxes, permits, and employee requirements.

The company will adopt a Line and Staff organizational structure, where the CEO and COO work as strategic partners at the top. The COO oversees day-to-day operations, directly managing the Lead Software Engineer, who leads the Development team including Aerodynamics and Flight Simulation Specialists. The Support branch, including Marketing, Community Managers, and Administrative Staff, ensures smooth operations and customer engagement.

The operating period begins with an 8-month preparation and development phase, followed by a 12-month launch and initial operation period, and later progressing into full-scale operations. The initial capital of **₱ 1,771,000** will be equally contributed by the partners and allocated for software development, office setup, marketing, and operational expenses.

Through this structure, AeroGenesis ensures clear authority, collaboration between top management, and efficient workflow across Development and Support teams, positioning the company for growth in the aviation simulation and technology market.

G. Capitalization

The investment goal for this project costs **₱ 1,771,000** for assets and equipment procurement thus:

Partner	Type of Partner Capitalist	Capital	Source of Capital
Rexus Bryan L. Gan	Capitalist	₱ 1452000	Personal Savings
Jaeden Balayo	Capitalist	₱ 1452000	Personal Savings

Table 2. Capitalization

H. Products Offered

Features/Services	**Pricing**
Game Flight Simulator/ VS mode	Free To Play
Aircraft Creation Mode	One Time Payment (₱499)
Addon Tools (For Optimization)	₱150 – ₱300
UGC Price Caps	
Light Aircraft Normal and Utility Category	₱ 50 – ₱60
Commercial Aircraft	120 – ₱250
Military Roles	
Interceptor	₱150 – ₱1000
Multi Role	₱300 – ₱1000
Fighter Jets	₱350 – ₱1000
Acrobatic Fighters (WW II)	₱120– ₱300
Stealth Bomber Aircraft	₱500 – ₱1000

Table 2.1 Products Offered

I. List of Personnel

Personnel	Working Hours / Compensation	Job Description	Job Specification
Game Developers (Senior Programmers)	**Working Hours:** 40 Hourse per Week **Monthly Salary:** 35,000	- Architect and implement core gameplay systems (movement, combat, AI, physics, UI logic). - Integrate art, audio, networking, and tools into a coherent build. - Optimize performance	- 5+ years professional game development experience. - Strong C++ and/or C# (Unreal / Unity respectively). - Experience shipping at least one commercial title (non-negotiable). - Familiarity with profiling tools and performance debugging.
Backend / Server Programmer	**Working Hours:** 40 Hourse per Week **Monthly Salary:** 35,000	- Build and maintain game servers (auth, matchmaking, persistence, leaderboards). - Handle concurrency, latency, cheating prevention, and crashes - Design APIs that won't collapse - Manage databases and server infrastructure.	- 3–5+ years backend development. - Experience with: REST / gRPC APIs SQL & NoSQL databases - Server scaling and load balancing - Understanding of networking fundamentals (TCP/UDP, latency, serialization). - Experience deploying to cloud platforms (AWS, GCP, Azure).
3D Modeler	**Working Hours:** 40 Hourse per Week **Monthly Salary:** 35,000	- Create characters, props, weapons, vehicles, and assets that	- 2–4+ years professional 3D art experience. - Proficiency in: • Blender / Maya / 3ds Max

		- Optimize topology, UVs, and LODs. - Work closely with animators and engine constraints.	• Substance Painter / Designer - Strong understanding of: • Game-ready topology • UV unwrapping • Normal maps & PBR workflow
UI/UX Developer	**Working Hours:** 40 Hourse per Week **Monthly Salary:** 35,000	- Design and implement game UI - Translate UX design into working, responsive in-game UI. - Ensure accessibility, readability, and controller compatibility.	- 2–4+ years UI/UX experience (games preferred). - Experience with: • Unity UI / Unreal UMG • Figma, Adobe XD, or similar - Solid understanding of: • User flow • Interaction feedback • Resolution scaling
Environment / Effects Artist	**Working Hours:** 40 Hourse per Week **Monthly Salary:** 35,000	- Build immersive worlds, lighting, atmospheres, and visual effects. - Create VFX (particles, explosions, weather, water, magic, etc.). - Balance visual quality with performance constraints.	- 2–5+ years environment or VFX art experience. - Strong in: • Unreal Niagara / Unity VFX Graph • Level composition and lighting - Understanding of: • Real-time rendering limitations • Shader basics
QA	**Working Hours:** 40 Hourse per Week **Monthly Salary:** 30,000	- Write clear, reproducible bug reports. - Verify fixes and prevent regressions.	- 1–3+ years QA or game testing experience. - Familiarity with:

		- Test performance, edge cases, and user flows.	• Bug tracking tools (Jira, TestRail, etc.) - Strong attention to detail and communication. - Understanding of basic game mechanics and player behavior.
Marketing	**Working Hours:** 40 Hourse per Week **Monthly Salary:** 25,000	- Manage social media, community, trailers, and store pages. - Coordinate launch campaigns and post-launch visibility.	- 2–5+ years marketing experience (games or digital products). - Experience with: • Social media campaigns • Influencer outreach • Steam / app store optimization - Basic analytics understanding (CTR, conversion, retention).
Admin	**Working Hours:** 40 Hourse per Week **Monthly Salary:** 25,000	- Manage Finance and industry Partnerships	• Office Tools & Software Setup • HR & Payroll Basics • Documentation & Record-Keeping • Internal Communication & Reporting

Table 2.2 List of Personnel

J. Employment Processes

Applicants for AeroGenesis must submit the following documents as part of their application:

Fully accomplished application form or resume

- 2 pieces of 2x2 colored pictures
- 2 pieces of 1x1 colored pictures
- NSO authenticated birth certificate (photocopy)
- SSS Number
- TIN Number
- NBI Clearance
- Police Clearance
- Drug Test Certificate

- CAAP/PRC License (for specialized positions such as Aerodynamics Specialists, Drone
- Operators, or other relevant roles)
- Medical Examination Report
- PhilHealth Number
- Pag-IBIG Num

Employment Process:

1. **Application Submission**

   Applicants must submit their application form or resume along with the required documents to the COO.

2. **Resume Review & Screening**

   The COO will review the application and analyze the resume to ensure that the applicant meets the initial requirements

3. **Interview & Technical Assessment**

   Shortlisted candidates will be invited for an interview and, if applicable, a technical assessment to evaluate their skills and suitability for the position. Technical roles, such as software development or aerodynamics specialists, will undergo practical tests.

4. **Selection & Job Offer**

   Candidates who pass the assessment and interview will receive a formal job offer detailing their position, responsibilities, and compensation.

5. **Orientation & Training**

   New hires will undergo an orientation and training program to familiarize them with company policies, workflow, software tools, project standards, and safety protocols.

6. **Regular Performance Evaluation**

   Employee performance will be monitored through feedback,

evaluations, and periodic assessments to ensure alignment with company objectives and quality standards.

7. **Probationary Period**

   Employees undergo a probationary period to evaluate their performance, adaptability, and integration within their team. The COO, along with department heads, monitors progress and provides feedback.

8. **Regular Performance Monitoring and Evaluation**

   After probation, employees are regularly evaluated through performance appraisals, feedback sessions, and KPI tracking to ensure productivity, quality, and alignment with company objectives.

9. **Continuous Training and Development**

   AeroGenesis provides ongoing professional development programs, workshops, and training sessions to upgrade employee skills, improve performance, and support career growth within the company.

K. TRAINING PROGRAM

Personnel	# of Days	Location	Trainings
All Roles	2 Days	DAVAO	• Studio Orientation • Introduction to Engine (Unreal/Unity/Custom) • Version Control Overview • Build & Deployment Overview • Mental-Model Crash Course
Role Specific Trainings			
Lead Engineer	3 Days	DAVAO	• Engine Architecture • Gameplay Systems Integration • Code Review Training • Team Coordination Exercises • Performance Profiling
Gameplay Physics Programmer	3 Days	DAVAO	• Physics Engine Fundamentals • Hands-On Coding Exercises • Debugging Physics Issues • Optimization Techniques • Physics Feature Project
Game Mechanics Programmer	3 Days	DAVAO	• Core Gameplay Loops • Scripting Interactive Systems • Parameter Tuning Exercises • Asset & UI Integration • Gameplay Feature Project
Backend / Server Programmer	3 Days	DAVAO	• Network Architecture • Database Integration • Security Basics • Server Scripting Practice • Stress Testing

Aircraft Modeler Developer	3 Days	DAVAO	• Aircraft Asset Pipeline • Physics-Aware Modeling • LOD, UV, Texturing • Rigging Basics • Engine-Ready Aircraft Project
3D Modeler	3 Days	DAVAO	• Tool Proficiency (Blender) • Asset Optimization • PBR Texturing Workflow • Engine Import & Materials • Prop/Character Project
UI/UX Developer	3 Days	DAVAO	• UX Principles • UI Programming in Engine • Responsive Layouts • HUD/Menu Implementation • Testing & Iteration
Environment / Effects Artist	3 Days	DAVAO	• World Building • Particle Systems & Shaders • Optimization Techniques • Scene Project • Cross-Role Collaboration
Cross-Role Integration	3 Days	DAVAO	• Studio-Wide Mini-Project • Teamwork & Pipeline Practice • Integration Supervision • Daily Reviews & Iteration
QA	2 Days	DAVAO	• Testing Tools & Bug Tracking • Writing Test Cases • Manual Playtesting • Reporting & Communication • Final QA Simulation / Mini Project

Marketing	2 Days	DAVAO	• Target Audience & Market Analysis • Marketing Tools & Campaign Planning • Social media & Community Engagement Practice
Admin	2 Days	DAVAO	• Office Tools & Software Setup • HR & Payroll Basics • Documentation & Record-Keeping • Internal Communication & Reporting

Table 2.3 Training Programs

L. Facilities, Machines & Equipment

a. FACILITIES

i. Office Area

ii. Equipment & Document Storage

iii. Utility Area

b. MACHINES

i. Printer

ii. Inverter Air Conditioners

iii. Computer Set

iv. Projector

v. Refrigerator

vi. Company vehicles

c. EQUIPMENTS

i. SD Card

ii. Hard Drive

iii. Top Box

iv. Tables and Chairs

v. Office Supplies

vi. Fire Extinguishers

M. Fringe Benefits

Salaries And Wages

AeroGenesis provides competitive salaries based on job roles and industry standards. Employees may be required to work beyond their regular hours due to business needs, but the company ensures a fair and reasonable workload to maintain work-life balance.

Overtime Pay

Any overtime work must be approved by the COO before it is rendered. Employees must submit their overtime records within the pay period to receive compensation. No approved overtime request, no overtime pay.

Rest Days And Holiday Pay

Employees are entitled to a 24-hour rest period after every six (6) consecutive working days. The following are considered regular holidays, during which employees will receive additional compensation if required to work:

▪ New Year's Day	January 01
▪ Maundy Thursday	Movable date
▪ Good Friday	Movable date
▪ Araw ng Kagitingan	April 09
▪ Labor Day	May 01
▪ Independence Day	June 12
▪ National Heroes Day	Last Sunday of August
▪ Eid'l Fitr	Movable date
▪ Eid'l Adha	Movable date
▪ Bonifacio Day	November 30
▪ Christmas Day	December 25

▪ Rizal Day December 30	Special holidays include:
▪ All Saints' Day	November 01
▪ Christmas Eve	December 24
▪ Last day of the Year	December 31

Employees working on a regular holiday will receive 100% of their daily rate, while work performed on a special holiday or rest day will be paid an additional 30% of their regular salary.

13th Month Pay

Employees shall receive their 13th-month pay, which is equivalent to 1/12 of their total basic salary earned within the year, provided no later than December 24 each year.

Year-End Bonuses

Employees may receive a year-end bonus, which serves as an additional incentive based on company performance, employee productivity, or profitability. This bonus is given at the discretion of the company and aims to reward employees for their contributions throughout the year.

Paternity And Maternity Leave

Eligible female employees are entitled to paid maternity leave, while male employees may avail of paternity leave upon proper documentation and compliance with company requirements.

Government-Mandated Benefits (Sss, Philhealth, Pag-Ibig)

AeroGenesis utions ensures that all employees receive comprehensive government-mandated benefits to support their financial security, healthcare needs, and long-term stability. These include:

- **Social Security System (SSS):** Provides financial assistance through retirement, disability, sickness, maternity, and death benefits, ensuring employees and their families are supported during crucial life events.
- **PhilHealth:** Covers medical expenses, hospitalization, outpatient services, and preventive healthcare, reducing the financial burden of healthcare costs for employees and their dependents.
- **Pag-IBIG Fund**: Offers affordable housing loan programs, savings opportunities, and financial assistance, helping employees secure a home and build long-term financial stability.

These benefits are automatically provided to all employees upon hiring, with contributions made by both the company and the employee, in accordance with government regulations.

Health And Wellness Benefits

Regular employees will be enrolled in a hospitalization plan under a trusted Health Maintenance Organization (HMO) to cover medical expenses, consultations, and emergency care. Employees will also receive annual medical check-ups to promote overall health and well-being.

Accident And Life Insurance

To ensure employee security, AeroGenesis provides accident insurance coverage for work-related and non-work-related accidents. Also, life insurance will be provided for eligible employees to offer financial protection for their families.

Professional Development And Training

AeroGenesis prioritizes career growth and skills development. Employees will have access to free professional training and certifications relevant to their roles, such as financial training for the accountant, and Skill enhancement trainings.

Employee Incentives And Awards

To recognize and reward outstanding employees, AeroGenesis will provide performance-based incentives and awards such as:

- Employee of the Month
- Attendance and Punctuality Awards
- Performance Milestone Awards
- "Above and Beyond" Recognition
- Excellence in Service Award

Additional benefits, including travel allowances, meal benefits, and productivity bonuses, may be given based on performance and company discretion.

Retirement Benefits

Employees who have served the company for a significant period will be entitled to retirement benefits, including financial incentives and assistance in processing their retirement claims from SSS and other government agencies.

N. Company Uniforms



Figure 2.3 Marketing uniform A.



Figure 2.4 Marketing uniform B.

TASK	Jan-Feb 2028	Mar-Apr 2028	May-Jun 2028	Jul-Aug 2028	Sept-Oct 2028	Nov-Dec 2028
Strategic Planning & Goal Setting						
Office Workplace Setup Recruitment Notice						
Job Interviews						
Staff Training and Orientation						
System Architecture Logic Planning						
Development Phase						
Marketing Strategy & Public Engagement						
Operational Readiness & Launch Prep						

Table 2.4 summary of the Project aspect

Chapter III

MARKET AND ECONOMIC ASPECTS

Market research is essential in determining the demand for AeroGenesis' aircraft design and simulation platform. A business may face challenges in maximizing growth if it focuses on only a narrow segment of users. To ensure steady expansion and long-term profitability, AeroGenesis will provide an accessible yet realistic aircraft creation and flight simulation experience that combines entertainment and educational value. The platform will serve aviation enthusiasts, simulation gamers, engineering students, and academic institutions that require interactive tools for learning aerodynamics, aircraft systems, and design principles. By catering to both recreational players and the academe, AeroGenesis can build a diverse and sustainable user base, strengthen its position in the niche aviation market, and create long-term opportunities through its user-generated content ecosystem.

A. Market Analysis

Understanding the market is essential in evaluating a business's potential for growth and long-term sustainability. Many startups struggle in their early years due to insufficient market research and unclear positioning. To avoid these challenges, AeroGenesis will carefully analyze industry trends, evaluate competitors, and identify the needs of both entertainment users and academic institutions to ensure the platform remains relevant and in demand. Recent trends show a growing interest in technology-driven simulation, sandbox creation tools,

and interactive learning platforms. Traditional learning methods in aerodynamics and aircraft design often rely heavily on theory, which can limit student engagement and practical understanding. At the same time, many existing flight simulation games focus mainly on flying rather than allowing users to design and test their own aircraft.

AeroGenesis addresses these gaps by providing a platform where users can design, modify, and simulate aircraft using realistic physics, combining both creativity and technical learning. For gamers, this offers a deeper level of engagement through user-generated content and aircraft customization. For academic institutions, the platform serves as an interactive educational tool that helps students better understand aerodynamics, flight stability, and aircraft systems through hands-on experimentation. Additionally, the increasing integration of simulation tools in STEM and engineering education presents a strong opportunity for AeroGenesis to expand its presence in schools and training programs. By serving both aviation enthusiasts and the academe, the company positions itself in a unique niche that blends entertainment, engineering, and digital creativity.

B. Market Study

Marketing plays a crucial role in ensuring that AeroGenesis effectively reaches its target users and builds a strong, sustainable community. It involves strategies in promotion, pricing, and platform accessibility to meet the needs of both aviation enthusiasts and academic institutions. Since AeroGenesis introduces

an innovative combination of aircraft design, realistic flight simulation, and user-generated content, it is important to educate potential users about how the platform differs from traditional flight simulators and standard educational tools.

To effectively promote the platform, the company will implement digital marketing strategies such as social media campaigns, video demonstrations, developer updates, and community showcases that highlight user-created aircraft and realistic flight physics. Search engine optimization (SEO), gaming forums, and partnerships with content creators in the aviation and simulation communities will further expand visibility. For the academic sector, the company will establish partnerships with schools, colleges, and training institutions offering aviation, engineering, and STEM programs to integrate AeroGenesis as a supplementary learning tool.

In addition to online promotion, the company will participate in gaming conventions, tech expos, and educational fairs to demonstrate the platform's capabilities. Workshops, webinars, and campus presentations will help educators and students understand how AeroGenesis can support learning in aerodynamics, aircraft systems, and design principles. To accommodate different users, the platform will adopt flexible pricing models, including free access for core gameplay, optional paid creation tools, and institutional licensing packages for schools. By combining innovative technology, strategic marketing, and community-focused engagement, AeroGenesis aims to establish itself as a leading platform in the

niche market of aircraft simulation and interactive aviation learning, positioning the business for long-term growth and industry recognition.

Year	Estimated PC Gamers (Philippines)	Increase in Gamers	Potential Clients (1%)
2029	20,000,000	---	200,000
2030	21,000,000	1,000,000	210,000
2031	22,050,000	1,050,000	220,500
2032	23,152,500	1,102,500	231,525
2033	24,310,125	1,157,625	243,101

Table 3.1 Estimated Market Potential for AeroGenesis PC Gaming Application

Based Game Player Acquisition					
YEAR	**Estimated Users (A)**	**Growth Rate (B)**	**Projected Users (Yc)**	**Price (₱)**	**TOTAL SALES (₱)**
2029	8,000	0%	8,000	499	₱3,992,000
2030	8,000	5%	8,400	499	₱4,191,600
2031	8,400	5%	8,820	499	₱4,398,180
2032	8,820	5%	9,261	499	₱4,619,739
2033	9,261	5%	9,724	499	₱4,850,276

Table 3.2 Based Game Player Acquisition

Demand Projection For PGC									
YEAR	**Active Players**	**Creators (10%)**	**Items Sold per Creator**	**Item Price Range (₱)**	**Total Item Sales Range (₱)**	**Mean Item Sales (₱)**	**Platform Fee (25%)**	**AeroGenesis Item Revenue Range (₱)**	**Mean Revenue (₱)**
2029	8,000	800	2–3	100–1,000	160,000 – 2,400,000	1,280,000	25%	40,000 – 600,000	320,000
2030	8,400	840	2–3	100–1,000	168,000 – 2,520,000	1,344,000	25%	42,000 – 630,000	336,000
2031	8,820	882	2–3	100–1,000	176,400 – 2,646,000	1,411,200	25%	44,100 – 661,500	352,800
2032	9,261	926	2–3	100–1,000	185,200 – 2,778,000	1,481,600	25%	46,300 – 694,500	370,400
2033	9,724	972	2–3	100–1,000	194,400 – 2,916,000	1,555,200	25%	48,600 – 729,000	388,800

Table 3.3 Demand Projection For PGC

Demand Projection For DLC					
YEAR	**DLC Expansion**	**Development Cost (₱)**	**DLC Price (₱)**	**Projected Buyers**	**Estimated Profit (₱)**
2030	Expansion Pack 1	750,000	399	5,200	1,324,800
2031	Expansion Pack 2	800,000	399	5,600	1,434,400
2032	Expansion Pack 2B	800,000	399	5,800	1,460,200

2033	Expansion Pack 3	850,000	399	6,000	1,544,000

Table 3.4 Demand Projection For DLC

C. Factors considered in preparing the project

The factors considered in the preparation of AeroGenesis include market demand, competition, regulatory considerations, and resource availability. The team analyzed the growing interest in realistic flight simulation, sandbox creation systems, and interactive learning platforms to determine how AeroGenesis can meet both entertainment and educational needs. Existing competitors in the flight simulation and aircraft-building genre were also evaluated to identify gaps in the market, particularly the lack of platforms that combine realistic physics with user-generated aircraft design. This analysis helps position AeroGenesis as a unique alternative that integrates creativity, engineering concepts, and gameplay.

In addition, legal and ethical considerations related to digital content creation, intellectual property protection, and online platform operations were reviewed to ensure responsible and compliant business practices. Financial feasibility was also assessed, including the cost of software development, server infrastructure, and platform maintenance. Resource planning covered the acquisition of skilled developers, artists, and technical staff, as well as the tools and technologies needed to build and sustain the simulation engine and UGC ecosystem. These combined factors guided the overall structure of the business model and support the long-term sustainability of the project.

D. Potential Customers Of The Project

❖ Aviation enthusiasts and flight simulation gamers who enjoy realistic aircraft operation and customization.

❖ Creative players interested in designing, modifying, and sharing their own aircraft through a user-generated content system.

❖ Students in aeronautical engineering, aviation technology, AMT, and other STEM-related programs who need interactive tools to better understand aircraft design and aerodynamics.

❖ Educational institutions such as colleges, universities, and training centers looking for simulation platforms to support engineering and aviation instruction.

❖ Content creators, modders, and simulation communities who seek platforms that allow deeper creativity, experimentation, and community sharing.

❖ Aviation hobbyist groups and online simulation communities that value realistic physics and collaborative aircraft development.

E. Supply Analysis

The availability of advanced development tools, skilled technical personnel, and reliable server infrastructure is essential for the success of AeroGenesis. The company will invest in high-performance software development tools, game engines, physics simulation systems, and secure online servers to ensure smooth gameplay and stable user-generated content integration. Additionally, the project requires skilled programmers, 3D modelers, artists, and system administrators who can develop and maintain realistic aircraft physics, detailed models, and an efficient online marketplace.

To support long-term operations, the company will implement regular system maintenance, server monitoring, and software updates to keep the platform optimized and secure. Content moderation systems and intellectual property protection measures will also be established to manage user-generated aircraft designs responsibly. By ensuring the availability of the right technology, talent, and digital infrastructure, AeroGenesis can deliver a high-quality simulation experience while sustaining a reliable and scalable platform for its growing community.

Chapter IV

TECHNICAL ASPECT

A. Product Description

AeroGenesis is a digital aircraft design and flight simulation platform that specializes in realistic aerodynamics, interactive aircraft creation, and user-generated content integration. The platform serves aviation enthusiasts, simulation gamers, and academic institutions by providing an engaging and educational environment for designing, testing, and analyzing aircraft performance. Our system delivers accurate, physics-based flight simulations that allow users to detect design flaws, improve efficiency, and understand real-world aerodynamic principles.

By utilizing advanced simulation technology and an integrated online marketplace for user-created aircraft, AeroGenesis offers a more immersive, creative, and analytical experience compared to traditional flight simulators that focus only on operating pre-built aircraft. The platform ensures a detailed, interactive, and scalable environment that supports both entertainment and educational applications.

B. Days In One Year The Project Will Operate

AeroGenesis operates as a digital platform accessible to users 24 hours a day, 7 days a week. However, for operational and administrative purposes, the development and management team will follow a structured work schedule. The

company estimates that internal operations will run approximately **312 days per year**, operating from Monday to Saturday, with Sundays reserved for system maintenance, server optimization, updates, and team rest.

Month	Number of Working Days	Total Working Hours
January	26	208
February	24	192
March	26	208
April	26	208
May	26	208
June	25	200
July	26	208
August	26	208
September	25	200
October	26	208
November	25	200
December	20	160
TOTAL	**312 Days**	**2,496 Hours**

Table 4.1 Days In One Year The Project Will Operate

C. Type Of Building

AeroGenesis will operate from a rented 50-square-meter office space that will serve as the central hub for software development, administrative functions, marketing coordination, and technical support. The office will accommodate the development team, including programmers, 3D modelers, and designers, as well as provide space for meetings, planning sessions, and content review. The layout will be structured to support collaborative work while maintaining an organized and professional environment suitable for both creative and technical tasks.

Cost Of Construction (Office Renovation)

Office Renovation Cost of 50 sqm Area:

Materials:

- Partition walls and drywall installation (for development and admin areas)
- Ceiling installation
- Flooring materials and installation
- Wall repainting and interior finishes
- Electrical rewiring and lighting installation (suitable for high-performance PCs)
- Air conditioning installation (1–2 units for equipment cooling)

- Networking setup and cable management infrastructure
- Interior designer's fees
- Workers' wages (labor)

Total Renovation Cost = ₱120,000

Assets and Liabilities			
Machines & Equipment			
High-Spec PC (Dev Workstation)	**Quantity**	**Cost per Unit (₱)**	**Total Amount (₱)**
Mid-Spec PC (Support/Admin)	11	75,000	825,000
Server (Dedicated)	4	35,000	140,000
Printer	1	180,000	180,000
Drawing Tablet	1	15,000	15,000
Split Type Aircon	3	8,000	24,000
Projector	2	40,000	80,000
Network Setup (Routers, Switches, Cabling)	1	20,000	20,000
Office Tables	1	35,000	35,000
Cubicles (2 per set)	4	9,000	36,000
Office Chairs	5	13,000	65,000
Conference Table	15	2,000	30,000
Conference Chairs	1	18,000	18,000
Whiteboard	10	1,500	15,000
Shelves	1	7,000	7,000
Office Steel Cabinet	2	2,500	5,000
Office Sofa Set	2	9,500	19,000
Business Permit & Registration	1	15,000	15,000
Software Licenses (Game Engine, Tools, Security)	1	25,000	25,000
Office Renovation (50 sqm)	1	150,000	150,000
Total	1	120,000	120,000
Assets and Liabilities			
Machines & Equipment	**₱1,804,000**		

TABLE 4.2 INVESTMENT COST

Operating Expenses	2029 (₱)	2030 (₱)	2031 (₱)	2032 (₱)	2033 (₱)
Electricity	90,000	92,700	95,481	98,345	101,296
Water	4,000	4,120	4,244	4,371	4,502
Building Maintenance	12,000	12,360	12,731	13,113	13,506
Office & Dev Consumables	150,000	154,500	159,135	163,909	168,826
Internet Bill	20,000	20,600	21,218	21,855	22,510
Salaries and Benefits (Lean Team)	5,616,000	1,956,000	1,956,000	1,956,000	1,956,000
Office Rent	240,000	247,200	254,616	262,255	270,123
Software Subscription	250,000	257,500	265,225	273,182	281,377
Business Permit Renewal	4,500	4,635	4,774	4,917	5,064
Certifications & Licensing	2,500	2,575	2,652	2,732	2,814
TOTAL	6,389,000	2,752,190	2,776,076	2,800,679	2,826,018

TABLE 4.3 OPERATION PERIOD EXPENSE

Chapter V

FINANCIAL ANALYSIS

A. INCOME STATEMENT (5 YEARS

YEAR	2029	2030	2031	2032	2033
Base Game Sales (₱)	3,992,000	4,191,600	4,398,180	4,617,739	4,850,276
Item Revenue Range (₱)	40,000 – 600,000	42,000 – 630,000	44,100 – 661,500	46,300 – 694,500	48,600 – 729,000
Mean Item Revenue (₱)	320,000	336,000	352,800	370,400	388,800
DLC Revenue (₱)	0	1,324,800	1,434,400	1,460,200	1,544,000
Total Revenue Range (₱)	4,032,000 – 4,592,000	5,557,600 – 6,145,400	5,876,680 – 6,493,080	6,124,239 – 6,772,439	6,442,876 – 7,123,276
Mean Total Revenue (₱)	4,312,000	5,851,500	6,187,880	6,243,379	6,783,076
Total Expenses (₱)	4,372,000	2,752,190	2,776,076	2,800,679	2,826,018
Net Revenue Range (₱)	-340,000 – 220,000	2,805,410 – 3,393,210	3,100,604 – 3,717,004	3,323,560 – 3,971,760	3,616,858 – 4,297,258
Mean Net Revenue (₱)	-60,000	3,099,310	3,411,804	3,442,700	3,957,058
MCIT 2% (₱)	86,240	117,030	123,758	124,868	135,662
Net Profit after MCIT (₱)	-146,240	2,982,280	3,288,046	3,317,832	3,821,396

TABLE 5.1.1 INCOME STATEMENT (5 YEARS)

B. DEPRECIATION TABLE

Asset	Quantity	Cost per Unit (₱)	Total Cost (₱)	Useful Life (Years)	Salvage Value (₱)	Annual Depreciation (₱)
High-Spec PC (Dev Workstation)	11	75,000	825,000	5	100,000	145,000
Mid-Spec PC (Support/Admin)	4	35,000	140,000	5	20,000	24,000
Server (Dedicated)	1	180,000	180,000	5	20,000	32,000
Printer	1	15,000	15,000	5	2,000	2,600
Drawing Tablet	3	8,000	24,000	5	3,000	4,200
Split Type Aircon	2	40,000	80,000	10	10,000	7,000
Projector	1	20,000	20,000	5	2,000	3,600
Network Setup (Routers, Switches)	1	35,000	35,000	5	5,000	6,000
Office Tables	4	9,000	36,000	10	4,000	3,200
Cubicles (2 per set)	5	13,000	65,000	10	10,000	5,500
Office Chairs	15	2,000	30,000	5	3,000	5,400
Conference Table	1	18,000	18,000	10	2,000	1,600
Conference Chairs	10	1,500	15,000	5	2,000	2,600
Whiteboard	1	7,000	7,000	5	1,000	1,200
Shelves	2	2,500	5,000	10	500	450
Office Steel Cabinet	2	9,500	19,000	10	1,500	1,750
Office Sofa Set	1	15,000	15,000	5	2,000	2,600

TABLE 5.1.2 DEPRECIATION TABLE

C. Balance Sheet

Year	2029	2030	2031	2032	2033
Cash / Bank	-77,000	3,100,210	3,409,304	3,649,660	3,957,058
Fixed Assets (Net of Depreciation)	1,151,600	634,200	116,800	0	0
Total Assets	1,074,600	3,734,410	3,526,104	3,649,660	3,957,058
Liabilities	0	0	0	0	0
Total Equity	1,074,600	3,734,410	3,526,104	3,649,660	3,957,058
Equity Owner A	537,300	1,867,205	1,763,052	1,824,830	1,978,529
Equity Owner B	537,300	1,867,205	1,763,052	1,824,830	1,978,529
Capital Owner A	537,300	1,867,205	1,763,052	1,824,830	1,978,529
Capital Owner B	537,300	1,867,205	1,763,052	1,824,830	1,978,529
Total Capital	1,074,600	3,734,410	3,526,104	3,649,660	3,957,058

TABLE 5.1.3 Balance Sheet

D. Statement of Cash flow

Year	Cash Inflow	Cash Outflow (Operating Expenses)	Net Flow from Operating Activities	Cash flows for Equipment Purchase	Cash Inflow (Owner's Equity Contribution)	Net Cash Flow from Financing Activities	Ending Cash Balance
2028	0	0	0	2,904,000	2,904,000	2,904,000	0
2029	4,312,000	6,389,000	-2,077,000	0	0	0	-2,077,000
2030	5,852,400	2,752,190	3,100,210	0	0	0	1,023,210
2031	6,185,380	2,776,076	3,409,304	0	0	0	4,432,514
2032	6,450,339	2,800,679	3,649,660	0	0	0	8,082,174
2033	6,783,076	2,826,018	3,957,058	0	0	0	12,039,232

TABLE 5.1.4 Statement of Cash flow

E. Return On Investment

Year	Net Profit After MCIT (₱)	Initial Investment (₱)	ROI (%)
2029	-2,035,460	2,904,000	-70.1%
2030	3,037,324	2,904,000	104.6%
2031	3,343,568	2,904,000	115.1%
2032	3,373,846	2,904,000	116.1%
2033	3,877,917	2,904,000	133.6%

TABLE 5.1.5 RETURN ON INVESTMENT

F. Return On Equity

Year	Net Profit After MCIT (₱)	Total Equity (₱)	ROE (%)	ROE Owner A (%)	ROE Owner B (%)
2029	-2,035,460	2,904,000	-70.1%	-35.1%	-35.1%
2030	3,037,324	5,941,324	51.1%	25.5%	25.5%
2031	3,343,568	9,284,892	36.0%	18.0%	18.0%
2032	3,373,846	12,658,738	26.6%	13.3%	13.3%
2033	3,877,917	16,536,655	23.5%	11.8%	11.8%

TABLE 5.1.6 RETURN ON EQUITY

G. Cash Payback Period

Year	Net Cash Flow from Operations (₱)	Cumulative Cash Flow (₱)
2028	-2,904,000 (Equipment Purchase)	-2,904,000
2029	-2,035,460	-4,939,460
2030	3,037,324	-1,902,136
2031	3,343,568	1,441,432
2032	3,373,846	4,815,278
2033	3,877,917	8,693,195

TABLE 5.1.7 CASH PAYBACK PERIOD

H. Break Even

Year	Fixed Costs (₱)	Variable Cost / Unit (₱)	Selling Price / Unit (₱)	Contribution Margin / Unit (₱)	Break-Even Units	Break-Even Sales (₱)	Fixed Costs (₱)
2029	6,389,000	150	499	349	**18,306 units**	9,134,694	6,389,000
2030	2,752,190	150	499	349	**7,886 units**	3,934,114	2,752,190
2031	2,776,076	150	499	349	**7,954 units**	3,968,046	2,776,076
2032	2,800,679	150	499	349	**8,024 units**	4,003,976	2,800,679
2033	2,826,018	150	499	349	**8,097 units**	4,040,403	2,826,018

TABLE 5.1.8 Break Even

PRODUCT DESCRIPTION

AeroGenesis is a PC-based gaming application designed to provide players with an immersive and interactive aviation-themed gaming experience. The game combines strategic gameplay, customizable aircraft systems, and community-driven content to create a dynamic and evolving digital platform for aviation and gaming enthusiasts.

The application allows players to explore various aviation environments, engage in mission-based gameplay, and customize aircraft through a wide range of in-game modifications and visual enhancements. The game is developed to run on personal computers and aims to deliver optimized performance, user-friendly controls, and engaging gameplay mechanics suitable for both casual and experienced players.

Salaries (Support & Operations)

After the game is launched, the development team is reduced to a small lean team responsible for system maintenance and operations. This team focuses on server monitoring, quality assurance, bug fixes, and community support. As a result, salary expenses decrease significantly compared to the full development phase

Server and Hosting

Server and hosting expenses cover the infrastructure required to maintain the game’s multiplayer functionality and online services. These costs include maintaining servers, handling player data, ensuring stable gameplay connections, and implementing system updates when necessary.

A. Income Statement (5 Years)

YEAR	2029	2030	2031	2032	2033
Base Game Sales (₱)	3,992,000	4,191,600	4,398,180	4,617,739	4,850,276
Item Revenue Range (₱)	40,000 – 600,000	42,000 – 630,000	44,100 – 661,500	46,300 – 694,500	48,600 – 729,000
Mean Item Revenue (₱)	320,000	336,000	352,800	370,400	388,800
DLC Revenue (₱)	0	1,324,800	1,434,400	1,460,200	1,544,000
Total Revenue Range (₱)	4,032,000 – 4,592,000	5,557,600 – 6,145,400	5,876,680 – 6,493,080	6,124,239 – 6,772,439	6,442,876 – 7,123,276
Mean Total Revenue (₱)	4,312,000	5,851,500	6,187,880	6,243,379	6,783,076
Total Expenses (₱)	4,372,000	2,752,190	2,776,076	2,800,679	2,826,018
Net Revenue Range (₱)	-340,000 – 220,000	2,805,410 – 3,393,210	3,100,604 – 3,717,004	3,323,560 – 3,971,760	3,616,858 – 4,297,258
Mean Net Revenue (₱)	-60,000	3,099,310	3,411,804	3,442,700	3,957,058
MCIT 2% (₱)	86,240	117,030	123,758	124,868	135,662
Net Profit after MCIT (₱)	-146,240	2,982,280	3,288,046	3,317,832	3,821,396

TABLE 5.2.1 Income Statement (5 Years)

Rent and Utilities

Rent and utilities include the cost of maintaining office space as well as essential services such as electricity, water, and internet connectivity. Since the team size is smaller after launch, the company may reduce workspace requirements, which helps control operational costs.

Miscellaneous Licenses and Permits

This category covers necessary business permits, software licenses, and other administrative requirements needed to legally operate the company and maintain the game's systems. These expenses ensure compliance with regulations and continued access to essential software tools.

B. Depreciation Table

Asset	Quantity	Cost per Unit (₱)	Total Cost (₱)	Useful Life (Years)	Salvage Value (₱)	Annual Depreciation (₱)
High-Spec PC (Dev Workstation)	11	75,000	825,000	5	100,000	145,000
Mid-Spec PC (Support/Admin)	4	35,000	140,000	5	20,000	24,000
Server (Dedicated)	1	180,000	180,000	5	20,000	32,000
Printer	1	15,000	15,000	5	2,000	2,600
Drawing Tablet	3	8,000	24,000	5	3,000	4,200
Split Type Aircon	2	40,000	80,000	10	10,000	7,000
Projector	1	20,000	20,000	5	2,000	3,600
Network Setup (Routers, Switches)	1	35,000	35,000	5	5,000	6,000
Office Tables	4	9,000	36,000	10	4,000	3,200

Cubicles (2 per set)	5	13,000	65,000	10	10,000	5,500
Office Chairs	15	2,000	30,000	5	3,000	5,400
Conference Table	1	18,000	18,000	10	2,000	1,600
Conference Chairs	10	1,500	15,000	5	2,000	2,600
Whiteboard	1	7,000	7,000	5	1,000	1,200
Shelves	2	2,500	5,000	10	500	450
Office Steel Cabinet	2	9,500	19,000	10	1,500	1,750
Office Sofa Set	1	15,000	15,000	5	2,000	2,600

Table 5.2.2 Depreciation Table

C. Total Operating Cost

The total estimated post-launch operating cost is approximately ₱3,000,000 per year. This reflects a significant reduction compared to the development phase, primarily due to the smaller operational team and lower development-related expenses

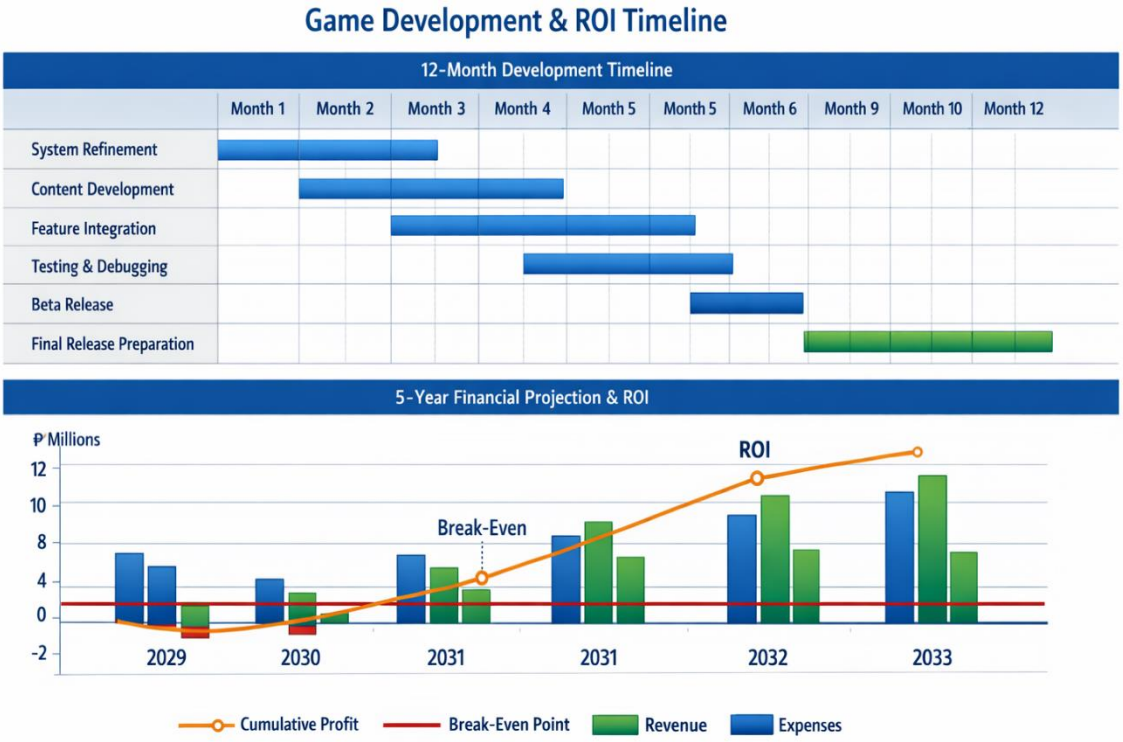


Figure 3 Total Operating Cost

Break-Even Timeline Projection - Assuming moderate player growth.

YEAR	ESTIMATED PLAYERS	GAME SALES	REVENUE	PROFIT / LOSS
2029	8,000	8,000	₱2,794,400	-₱1,577,600
2030	13,000	13,000	₱4,540,900	₱168,900
2031	18,000	18,000	₱6,287,400	₱1,915,400
2032	24,000	24,000	₱8,383,200	₱4,011,200
2033	30,000	30,000	₱10,479,000	₱6,107,000

Table 5.2.3 Break-Even

2029 – Full Development Phase

In 2029, the project focuses on full game development, including system completion, content creation, feature integration, testing, and launch preparation. The total operating expense is estimated at ₱4,372,000. With approximately 12,520 copies sold, the project reaches its break-even point during the launch year, meaning all revenue covers development costs and no profit is generated yet.

2030 – Maintenance and Minor Updates

In 2030, the project transitions to maintenance mode with reduced development activities focused on bug fixes, system maintenance, and community management. Operating costs decrease due to a smaller team and fewer development requirements. With estimated sales of 13,000 copies, the project generates its first profit of about ₱1,211,913

2031 – Maintenance and DLC Expansion

During 2031, the game continues in maintenance mode while introducing DLC expansions to maintain player engagement and generate additional revenue. With approximately 18,000 copies sold, the project earns around ₱2,004,812 in profit, bringing cumulative profit closer to achieving return on investment.

2032 – Content Expansion and Events

In 2032, new maps, events, and additional content are introduced to expand the player base and maintain engagement. With about 24,000 copies sold, the project generates an annual profit of approximately ₱3,348,192 and achieves its return on investment, surpassing the initial development cost.

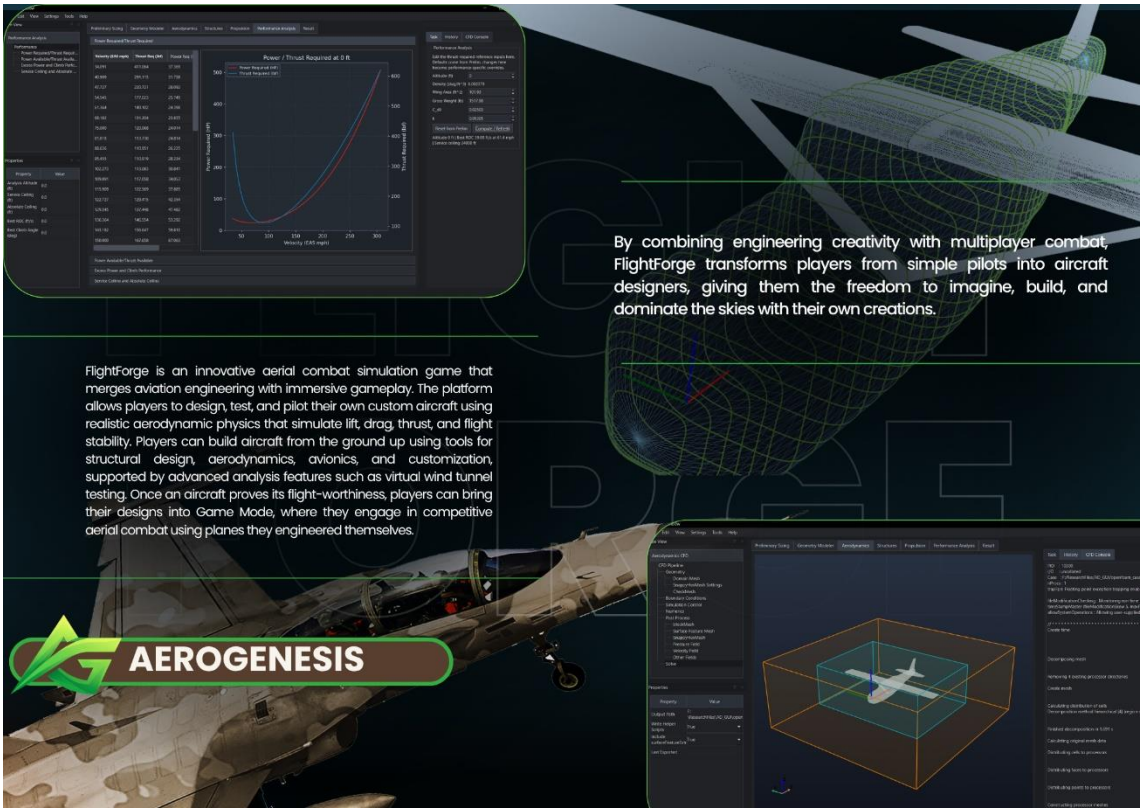
2033 – Competitive Events and Community Growth

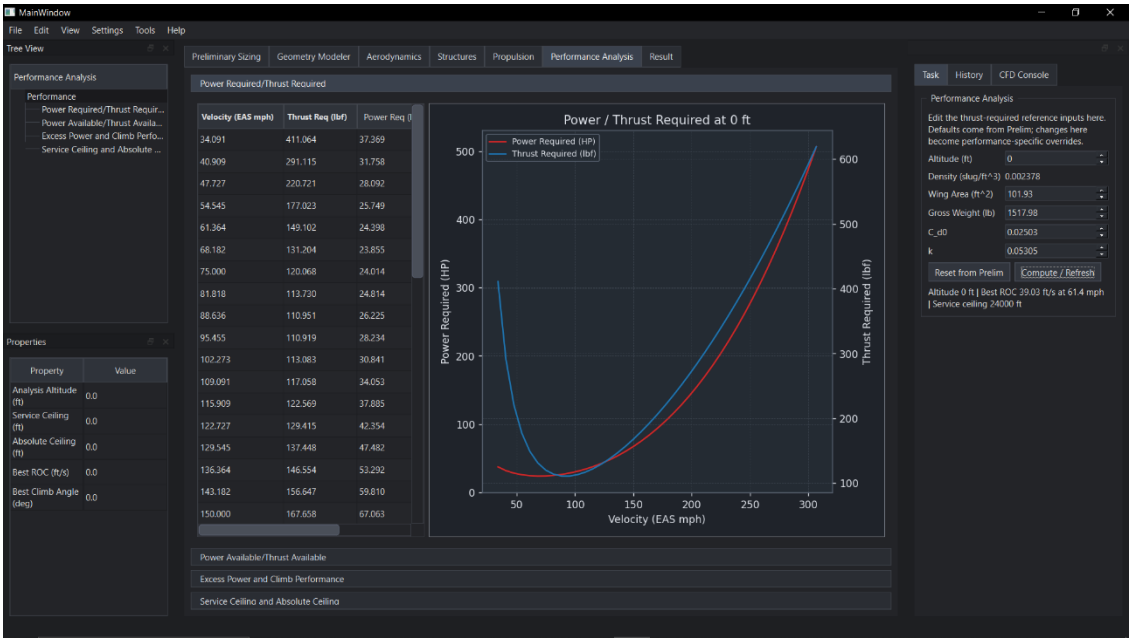
By 2033, the game reaches a mature stage focused on community growth, competitive events, and continued content support. With projected sales of 30,000 copies, the project generates about ₱5,095,212 in annual profit, demonstrating strong financial growth and long-term sustainability.

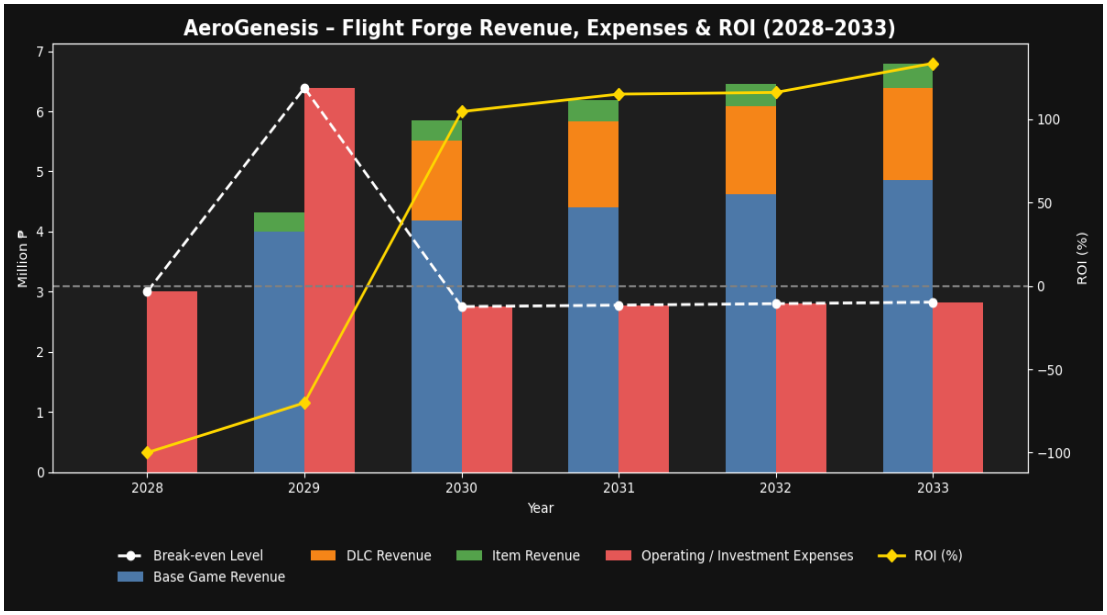
APENDIX A

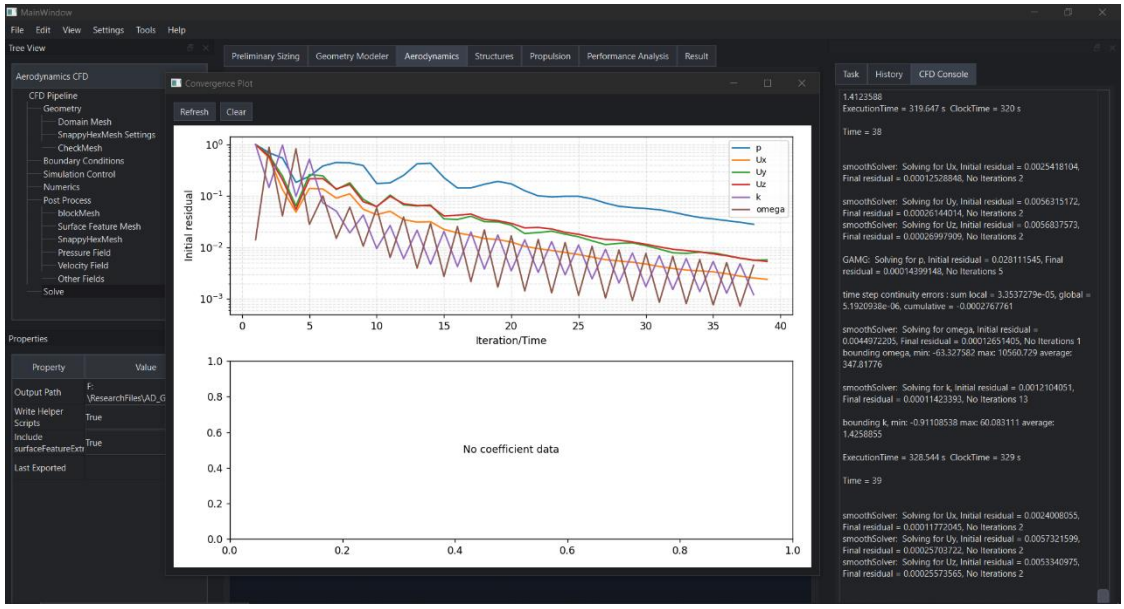

25m
BREAK ROOM
LOUNGE AREA
DEVELOPERS ROOM
PRINTING AREA
SERVER ROOM
M
V
W
OPEN WORK AREA
RECEPTION
CONFERENCE ROOM

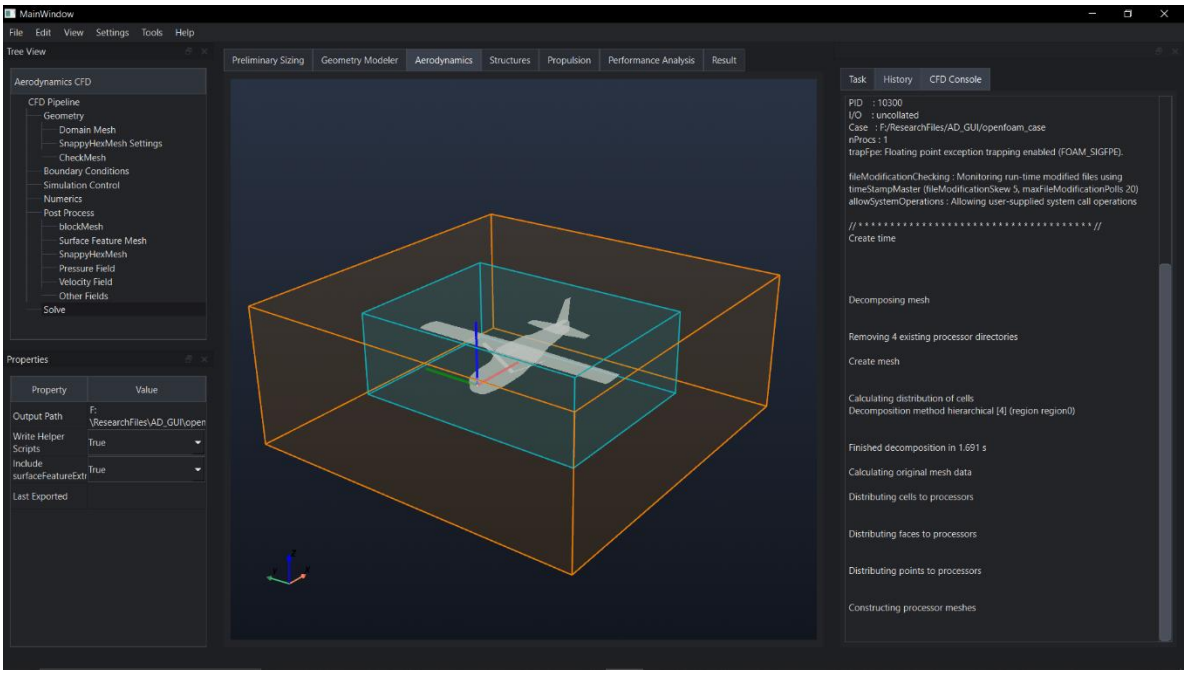

FlightForge
BUILD AND PLAY
www.reallygreatsite.com
AEROGENESIS

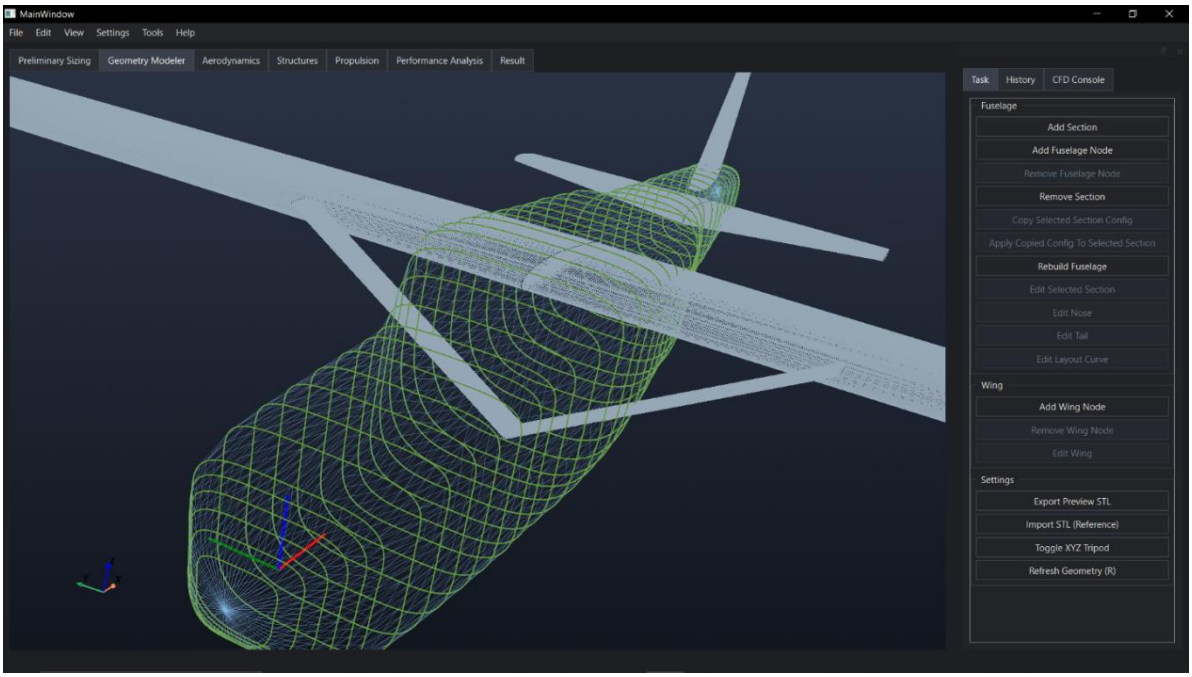

By combining engineering creativity with multiplayer combat, FlightForge transforms players from simple pilots into aircraft designers, giving them the freedom to imagine, build, and dominate the skies with their own creations.
FlightForge is an innovative aerial combat simulation game that merges aviation engineering with immersive gameplay. The platform allows players to design, test, and pilot their own custom aircraft using realistic aerodynamic physics that simulate lift, drag, thrust, and flight stability. Players can build aircraft from the ground up using tools for structural design, aerodynamics, avionics, and customization, supported by advanced analysis features such as virtual wind tunnel testing. Once an aircraft proves its flight-worthiness, players can bring their designs into Game Mode, where they engage in competitive aerial combat using planes they engineered themselves.
AEROGENESIS


MainWindow
File Edit View Settings Tools Help
Tree View
Performance Analysis
Performance
Power Required/Thrust Requir...
Power Available/Thrust Availa...
Excess Power and Climb Perfo...
Service Ceiling and Absolute ...
Properties
Property Value
Analysis Altitude (ft) 0.0
Service Ceiling (ft) 0.0
Absolute Ceiling (ft) 0.0
Best ROC (ft/s) 0.0
Best Climb Angle (deg) 0.0
Preliminary Sizing Geometry Modeler Aerodynamics Structures Propulsion Performance Analysis Result
Power Required/Thrust Required
Velocity (EAS mph) Thrust Req (lbf) Power Req (
34.091 411.064 37.369
40.909 291.115 31.758
47.727 220.721 28.092
54.545 177.023 25.749
61.364 149.102 24.398
68.182 131.204 23.855
75.000 120.068 24.014
81.818 113.730 24.814
88.636 110.951 26.225
95.455 110.919 28.234
102.273 113.083 30.841
109.091 117.058 34.053
115.909 122.569 37.885
122.727 129.415 42.354
129.545 137.448 47.482
136.364 146.554 53.292
143.182 156.647 59.810
150.000 167.658 67.063
Power / Thrust Required at 0 ft
Power Required (HP)
Thrust Required (lbf)
Power Required (HP)
Thrust Required (lbf)
Velocity (EAS mph)
Power Available/Thrust Available
Excess Power and Climb Performance
Service Ceiling and Absolute Ceiling
Task History CFD Console
Performance Analysis
Edit the thrust-required reference inputs here. Defaults come from Prelim; changes here become performance-specific overrides.
Altitude (ft) 0
Density (slug/ft^3) 0.002378
Wing Area (ft^2) 101.93
Gross Weight (lb) 1517.98
C_d0 0.02503
k 0.05305
Reset from Prelim
Compute / Refresh
Altitude 0 ft | Best ROC 39.03 ft/s at 61.4 mph | Service ceiling 24000 ft


AeroGenesis - Flight Forge Revenue, Expenses & ROI (2028-2033)
Million ₱
ROI (%)
Year
2028
2029
2030
2031
2032
2033
Break-even Level
Base Game Revenue
DLC Revenue
Item Revenue
Operating / Investment Expenses
ROI (%)


Preliminary Sizing
Geometry Modeler
Aerodynamics
Structures
Propulsion
Performance Analysis
Result
Convergence Plot
Refresh
Clear
Initial residual
Iteration/Time
p
Ux
Uy
Uz
k
omega
No coefficient data
Task
History
CFD Console


MainWindow
File Edit View Settings Tools Help
Tree View
Aerodynamics CFD
CFD Pipeline
Geometry
Domain Mesh
SnappyHexMesh Settings
CheckMesh
Boundary Conditions
Simulation Control
Numerics
Post Process
blockMesh
Surface Feature Mesh
SnappyHexMesh
Pressure Field
Velocity Field
Other Fields
Solve
Properties
Property
Value
Output Path
Write Helper Scripts
Include surfaceFeatureExtr
Last Exported
Preliminary Sizing
Geometry Modeler
Aerodynamics
Structures
Propulsion
Performance Analysis
Result
Task
History
CFD Console
PID : 10300
I/O : uncollated
Case : F:/ResearchFiles/AD_GUI/openfoam_case
nProcs : 1
trapFpe: Floating point exception trapping enabled (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using timeStampMaster (fileModificationSkew 5, maxFileModificationPolls 20)
allowSystemOperations : Allowing user-supplied system call operations
Create time
Decomposing mesh
Removing 4 existing processor directories
Create mesh
Calculating distribution of cells
Decomposition method hierarchical [4] (region region0)
Finished decomposition in 1.691 s
Calculating original mesh data
Distributing cells to processors
Distributing faces to processors
Distributing points to processors
Constructing processor meshes


MainWindow
File Edit View Settings Tools Help
Preliminary Sizing
Geometry Modeler
Aerodynamics
Structures
Propulsion
Performance Analysis
Result
Task
History
CFD Console
Fuselage
Add Section
Add Fuselage Node
Remove Fuselage Node
Remove Section
Copy Selected Section Config
Apply Copied Config To Selected Section
Rebuild Fuselage
Edit Selected Section
Edit Nose
Edit Tail
Edit Layout Curve
Wing
Add Wing Node
Remove Wing Node
Edit Wing
Settings
Export Preview STL
Import STL (Reference)
Toggle XYZ Tripod
Refresh Geometry (R)

APENDIX B

DEPRECIATION

$$Annual\ Depreciation = \frac{Cost - Salvage\ Value}{Useful\ Life}$$

Asset	Quantity	Cost per Unit (₱)	Total Cost (₱)	Estimated Life (Years)	Salvage Value (₱)
Printer	2	₱ 13,395.00	₱ 26,790.00	5	₱2,679.00
Split type Aircon	7	₱ 40,370	₱ 80,740.00	5	₱8,074.00
Computer Set	16	₱ 20,000.00	₱ 240,000.00	5	₱24,000.00
Projector	1	₱ 20,000.00	₱ 20,000.00	5	₱2,000.00
SD Card	4	₱ 700.00	₱ 2,800.00	5	₱280.00
Hard drive	4	₱ 2,000.00	₱ 8,000.00	5	₱800.00
Motorcycles	4	₱ 79,000.00	₱ 316,000.00	5	₱31,600.00
Top Box	4	₱ 4,500.00	₱ 18,000.00	5	₱1,800.00
Small Refrigerator	1	₱ 8,000.00	₱ 8,000.00	5	₱800.00

RETURN ON INVESTMENT

$$ROI = \left(\frac{Total\ Profit\ after\ tax}{Total\ Investment\ Cost}\right) \times 100$$

Year	Net Profit (₱)	Total Investment (₱)	ROI (%)
2029	4,312,000	2,904,000	148.5%
2030	5,852,400	2,904,000	201.5%
2031	6,185,380	2,904,000	213.0%
2032	6,450,339	2,904,000	222.0%
2033	6,783,076	2,904,000	233.5%

CASH PAYBACK PERIOD

$$CPP = \frac{Total\ Investment\ Cost}{Average\ Net\ Profit}$$

Year	Net Cash Flow from Operations (₱)	Cumulative Cash Flow (₱)
2028	-2,904,000 (Equipment Purchase)	-2,904,000
2029	-2,077,000	-4,981,000
2030	3,037,324	-1,943,676
2031	3,343,568	1,399,892
2032	3,373,846	4,773,738
2033	3,877,917	8,651,655

BREAK-EVEN POINT

$$B.E.P = \frac{Total\ Fixed\ Cost}{1 - \frac{Total\ Variable\ Cost}{}}$$

Year	Base Game Revenue (₱)	DLC Revenue (₱)	Item Revenue (Mean ₱)	Total Revenue (₱)	Operating Expenses (₱)	Contribution Margin (₱)	BEP Revenue (₱)	BEP Units
2029	3,992,000	0	320,000	4,312,000	6,389,000	-2,077,000	N/A	N/A
2030	4,191,600	1,324,800	336,000	5,851,500	2,752,190	3,099,310	2,752,190	5,516
2031	4,398,180	1,434,400	352,800	6,187,880	2,776,076	3,411,804	2,776,076	5,566
2032	4,617,739	1,460,200	370,400	6,243,379	2,800,679	3,442,700	2,800,679	5,613
2033	4,850,276	1,544,000	388,800	6,783,076	2,826,018	3,957,058	2,826,018	5,666

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CURRICULUM VITAE



PERSONAL BACKGROUND

Name	**:** Jaeden Lloyd D. Balayo
Address	**:** Suterville, San Jose Gusu, Zamboanga City
Birth Date	**:** August 7, 2003
Civil Status	**:** Single
Father's Name	**:** Abraham A. Balayo
Mother's Name	**:** Ille Joy D. Balayo

EDUCATIONAL BACKGROUND

Elementary	**:** J-Jireh School Inc.
Junior High School	**:** J-Jireh School Inc.
Senior High School	**:** J-Jireh School Inc.
College (on going)	**:** MATS College of Technology

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Telephone No. 082-224-7928

CURRICULUM VITAE



PERSONAL BACKGROUND

Name : Rexus Bryan L. Gan

Address : Tamisan, Mati City
Birth Date : July 31, 2002

Civil Status : Single

Father's Name : Rafael S. Gan

Mother's Name : Minviluz Gan

EDUCATIONAL BACKGROUND

Elementary : Bobon Elementary School

Junior High School : Mati National Comprehensive High School

Senior High School : Mati Doctors Academy

College (on going) : MATS College of Technology